HOWARD RELEASE NOTES

1 RELEASE NOTES

1.1 0.14.0

This release introduces several significant new features and improvements, primarily focused on enhancing annotation capabilities and overall performance. Key additions include a new BigWig/BigBed annotation method with aggregation, a fast strategy for Parquet annotation, and a chunking process to handle very large input files. This version also brings a new resource management strategy to optimize memory and speed. Updates focus on improving resource usage for annotation and calculation processes, ans external tools like snpEff and Annovar, while numerous fixes address issues with data downloads, VCF calculations, and transcript annotations.

1.1.1 News

- Add pipeline mode to define multiple ordered steps of operations to process (between "annotation", "calculation" and "prioritization" operation types) (#522)
- Add docker tools to extend annotation with external tools (such as VEP) (#527)
- Add and improve BigWig/BigBed annotation, with aggregation methods (#444)
- Add fast strategy for Parquet annotation (#466)
- Add chunking process for huge input files (#442)

1.1.2 Updates

- Improve annotation update from external tools (e.g. snpEff, Annovar, splice...), to improve ressources (#443)
- Add strategy to manage resources, based on CTAS method and duckDB spilling identification (#462, #466, #459)
- Improve Parquet annotation (add options), resources dealing, fast parameter (#466)
- Improve transcripts process (calculation), especially for transcript prioritization (#475, #492, #495)
- Improve calculation tool by adding named parameters on section (#503, #505), and add/improve some pertinent calculations (#496, #501, #519, , #519)
- Improve Annovar annotation by adding supplementary configuration sections, and a specific 'parallelize' option to speedup the annotation process (#508 and #550)
- Clarification and homogenization of annotations fields for databases annotation (Parquet method, as well as Annovar, BCFtools, snpSift and BigWig) (#514)

1.1.3 Fixes

- Fix new snpEff URL (#441)
- Fix Annovar check DB files (*_avdblist.txt) download (only for download, not when auto-download during annotation) (#440)
- Fix VCF calculation to consider '.' values (#445)
- Fix Alphamissense download (#448)
- Fix issue when some trancript annotations are missing (#451)
- Fix external tool update with REF/ALT in upper case (#451)
- Fix Aria2 threads limitation (#467)
- Fix transcript export (#475)
- Fix loading VCF with sample in integer format (#478)
- Fix region annotations with BED file (#487)
- Fix annovar download to prevent inappropriate update (#507)

1.2 0.13.0

Introduction of interactvie mode to dynamically explore data, using variants view with columns types (defined in VCF header), with build-in tool or Harlequin tool. Improvements using view duckDB feature, mainly for calculations and prioritizations, and transcripts mapping. New useful tools and plugins.

1.2.1 News

- Interactivity mode:
  - Interactive terminal to execute SQL queries
  - Completion and history features
  - Display modes (e.g. 'dataframe', 'markdown', 'simple', 'tabulate')
  - Query result limitation, to speedup exploration
  - Extended mode with Harlequin tool (https://harlequin.sh)
- Variants "view":
  - Query more easily (with "variants_view" view loaded by default in query tools)
  - INFO/tags and samples structured on specific columns ('INFOS' and 'SAMPLES')
  - Improve performances for annotations, calculations and prioritizations
  - Improve prioritization profiles definition using columns types (Integer, Float, String) with list type (defined in VCF header)
- New tools:
  - Tool 'filter' to filter variants in SQL format and samples
  - Tool 'sort' to sort variants from contig order
- Plugins:
  - 'to_excel': Convert VCF to Excel '.xlsx' format

1.2.2 Updates

- Improve prioritization performances (vith annotations view)
- Improve transcripts prioritization performances (reduce memory)
- Improve transcripts view creation (columns type detected)
- Improve snpEff extract information (HGVS list, explode annotations, generate as JSON)
- Improve NOMEN extraction from HGVS list, including multiple patterns for NOMEN
- Improve rename or remove INFO/tags
- Improve stats generation (speedup, annotations stats, custom queries)
- Improve genomes download with URL or local file as original FASTA
- Improve dbNSFP download with URL or local file
- Improve export in Parquet format by keeping column types

1.2.3 Fixes

- Fix BARCODEFAMILY calculation (multiple barcode with successive commandes)
- Fix dnSNP download and index generated vcf
- Fix explode_infos parameter
- Fix empty VCF fails
- Fix header generation

1.3 0.12.1.1

Few updates and fixes.

1.3.1 Updates

- Improve transcripts table creation
- Improve Docker image:
  - Using Micromamba instead of mamba
  - Reduce size by combining layers and cleaning caches

1.3.2 Fixes

- Fix tag of howard tool and docker images

1.4 0.12.0

This release introduce 'BigWig' annotation, prioritization options and transcripts view, improve samples managment, INFO/tags rename, annotation databases generation and operations, configuration files in YAML format, and python packages stability.

1.4.1 News

- Add annotation method 'BigWig'
- Add prioritization options:
  - SQL syntax available to define filters
  - New 'Class' prioritization field
- New transcripts view:
  - Create a transcript view, using a structure from multiple source type (e.g. snpEff, external annotation databases)
  - Mapping between multiple transcript ID source (e.g. refSeq, Ensembl)
  - Transcripts prioritization, using same prioritization process than variants
  - Export transcripts table as a file, in multiple format such as VCF, TSV, Parquet
- Export with a specific sample list
- Rename or remove INFO/tags before exporting
- Configuration and parameters files in YAML format allowed
- Add dynamic transcript column for NOMEN calculation (using transcript prioritization column)
- Add plugins:
  - 'update_databases'

1.4.2 Updates

- Improve snpEff annotations operations
- New option 'uniquify' for dbSNFP generation, identification of columns type
- Managment, check and export of samples columns
- Improve query type mode
- Improve splice annotation
- Improve NOMEN generation

1.4.3 Fixes

- Genotype format detection
- Fix packages releases
- Fix parameters and configuration files options
- Fix calculations list and parametrization
- Fix empty file export
- Fix BED annotation with parquet method
- More explicite log messages

1.5 0.11.0

This release introduce splice annotation tool, and update duckDB python package for improve stability.

1.5.1 News

- Add splice tool with docker image
- Add plugins:
  - 'genebe' (GeneBe annotation using REST API)
  - 'minimalize' (Minimalize a VCF file, such as removing INFO/Tags or samples)

1.5.2 Updates

- DuckDB 1.0.0 stable Snow Duck (Anas Nivis) release
- Add API Documentation
- Improve tests

1.5.3 Fixes

- Paths parameters check fixed (genome and genomes-folders)
- Fix snpEff download error with databases list

1.6 0.10.0

This release is a refactor of HOWARD (Highly Open Workflow for Annotation & Ranking toward genomic variant Discovery) in Python, using Parquet and duckDB.

HOWARD annotates and prioritizes genetic variations, calculates and normalizes annotations, translates files in multiple formats (e.g. vcf, tsv, parquet) and generates variants statistics.

See README and gitHub for more explanations.

1.7 Previous releases

See HOWARD gitHub for more information about previous releases.

1.7.1 0.9b-07/10/2016

- Script creation

1.7.2 0.9.1b-11/10/2016

- Add Prioritization and Translation

- Add snpEff annotation and stats

1.7.3 0.9.8b-21/03/2017

- Add Multithreading on Prioritization and Translation

1.7.4 0.9.9b-18/04/2017

- Add Calculation step

1.7.5 0.9.10b-07/11/2017

- Add generic file annotation through --annotation option
- No need to be in configuration file
- Need to be in ANNOVAR database folder (file 'ASSEMBLY_ANN.txt' for annotation 'ANN')
- Add options: --force , --split
- Add options for VCFanotation.pl: --show_annoataion, --show_annotations_full
- Add database download option nowget in VCFanotation.pl
- Fixes: multithreading, VAF calculation, configuration and check dependencies

1.7.6 0.9.11b-07/05/2018

- Replace VCFTOOLS command to BCFTOOLS command
- Release added into the output VCF
- Update SNPEff options
- Add VARTYPE, CALLING_QUALITY and CALLING_QUALITY_EXPLODE option on calculation
- Add description on calculations

1.7.7 0.9.11.1b-14/05/2018

- Improve VCF validation
- Fix snpEff annotation bug

1.7.8 0.9.11.2b-17/08/2018

- Add --vcf input vcf file option
- Create Output file directory automatically
- Improve Multithreading

1.7.9 0.9.12b-24/08/2018

- Improve Multithreading
- Input VCF compressed with BGZIP accepted
- Output VCF compression level
- Add VCF input sorting and multiallele split step (by default)
- Add VCF input normalization step with option --norm
- Bug fixes

1.7.10 0.9.13b-04/10/2018

- Multithreading improved
- Change default output vcf
- Input vcf without samples allowed
- VCF Validation with contig check
- Add multi VCF in input option
- Add --annotate option for BCFTOOLS annotation with a VCF and TAG (beta)
- Remove no multithreading part code to multithreading with 1 thread
- Remove --multithreading parameter, only --thread parameter to deal with multithreading
- Replace --filter and --format parameters by --prioritization and --translation parameters
- Add snpeff options to VCFannotation.pl

1.7.11 0.9.14b-21/01/2019

- Reorganization of folders (bin, config, docs, toolbox...).
- Improve Translation (TSV or VCF, sort on fields, selection of fields, filtering on fields), especially memory efficiency
- Change Number/Type/Description of new INFO/FORMAT header generated
- Remove snpEff option --snpeff and --snpeff_hgvs. SnpEff is used through --annotation option
- Add '#' to the TAB/TSV delimiter format header
- Update dbNSFP config annotation file script
- Change default configuration files for annotation (add dbSNFP 3.5a, update mcap and regspintron) and prioritization
- Bug fixed: file identification in annotation configuration
- Bug fixed: calculation INFO fields header, snpeff parameters options on multithreading
- Bug fixed: snpeff parameters in command line

1.7.12 0.9.15b-19/09/2019

- Rename HOWARD.sh to HOWARD.
- Add --nomen_fields parameter and update NOMEN calculation.
- Add --bcftools_stats and --stats parameter.
- Change PZScore, PZFlag, PZComment and PZInfos generation, adding default PZ and all PZ filters.
- Bug fixed: translation fields identification.

1.7.13 0.9.15.1b-28/05/2020

- Bug fixed: NOMEN calculation clear previous NOMEN values if using force option.

1.7.14 0.9.15.2-12/10/2020

- Change --norm parameter by adding '--check-ref=s' in bcftools command.
- Add --norm_options parameter.
- force translation VCF by default.
- Change VAF_stats and add DP_stats.

1.7.15 0.9.15.4-11/04/2021

- Remove --snpeff_threads parameter (for snpEff 5.0e compatibility) and improve --snpeff_stats.
- Add a check and rehead INFO fields if necessary (prevent some incorrect INFO header format).
- Fix --compress parameter and add --index parameter.

1.7.16 0.9.15.5-21/07/2021

- Add INFO description Type option, with autodetection.
- Add prioritization mode 'VaRank'/'max' for score calculation.
- Add calculation DP, AD, GQ and associated stats.
- Fix INFO field type for VCF.

1.7.17 0.9.15.6-16/09/2021

- Structural variant compatibility.
- NOMEN extraction and generation improved, with --nomen_pattern option (only for SNV and InDel).
- Improve translation with variant sorting.
- Improve error catching.